

Notes: Di Maggio & Yao (RFS, 2021)

Fintech Borrowers: Lax Screening or Cream-Skimming?

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1 Big picture

- **Question.** Do fintech lenders expand credit by serving borrowers that traditional banks fail to serve, or do they cream-skim the safest borrowers using better technology?
- **Core answer.** The evidence points to a mixed but distinctive fintech model. Fintech lenders initially enter markets by lending to riskier borrowers and later expand toward safer borrowers. They rely heavily on hard credit-report information and appear weaker at incorporating soft information.
- **Main data advantage.** The paper uses proprietary monthly borrower-level credit bureau data that identify lender names, separate fintech from traditional lenders, observe loan terms, and track all borrower liabilities before and after origination.
- **Main borrower-selection result.** Fintech borrowers are not simply hidden prime borrowers. They have many credit accounts, high credit utilization, and substantial access to credit; many appear to borrow for debt consolidation and additional consumption.
- **Main pricing result.** Fintech loans often charge higher rates to low-score and thin-file borrowers but offer better rates to high-score borrowers. The rate-default relationship is stronger for fintech loans, suggesting that fintech lenders price risk aggressively on the intensive margin.
- **Main performance result.** Fintech loans are more likely to become delinquent than comparable non-fintech loans, even after rich controls, zip-code-by-time fixed effects, matching, and borrower fixed effects.
- **Dynamic borrower outcome.** Fintech borrowers initially reduce some debt and improve credit scores, but debt balances later rebound and credit scores decline. They are more likely to finance durable consumption, such as car purchases.
- **Economic takeaway.** Fintech lenders expand consumer credit access and price risk with hard data, but the expansion comes with higher ex post borrower fragility. The key friction is not only access to credit; it is whether the lender can screen and discipline borrowers whose risk is partly soft-information-based.

2 Incremental contribution of the paper

- **Moves beyond single-platform evidence.** Earlier fintech lending studies often focused on a specific platform such as LendingClub or on application-level data. This paper uses credit bureau data covering multiple fintech and non-fintech lenders, which makes the results closer to the overall personal-loan market.
- **Creates a natural benchmark group.** The data include comparable borrowers receiving personal loans from traditional lenders in the same local markets and time periods. This lets the authors benchmark fintech borrowers against similar non-fintech borrowers rather than only against rejected applicants or platform participants.
- **Adds full-balance-sheet dynamics.** The paper does not stop at loan origination or default. It tracks credit scores, revolving debt, total debt, car purchases, repeat borrowing, and job-loss-related borrowing. This is important because fintech loans may initially look like debt consolidation but later generate higher leverage.
- **Separates hard-information pricing from soft-information screening.** The paper shows that observable credit-report variables explain more of fintech interest-rate variation than traditional-lender rate variation. This provides evidence that fintech lenders depend more on hard information and may have less local or relationship-based soft information.
- **Reconciles higher default with lender rationality.** Higher fintech default rates do not necessarily mean fintech lenders make mistakes. Their rates are strongly correlated with future defaults and repeat borrowing is high, implying that pricing and lifetime value may compensate for elevated default risk.

- **Contributes to household finance.** The paper shows how easier access to unsecured credit can change borrower behavior after origination. The relevant outcome is not merely loan approval or interest rate; it is the borrower’s subsequent debt accumulation and financial health.

3 Data and measurement

3.1 Credit bureau data and lender identification

The main data come from one of the largest U.S. credit bureaus. The bureau records monthly payment histories and tradeline information for many account types, including personal loans, credit cards, auto loans, student loans, mortgages, home equity lines, and other credit accounts. The proprietary version used in the paper is especially valuable because it identifies lender names, allowing the authors to classify loans by fintech and traditional institutions.

Interpretation: The key measurement contribution is lender observability. Without lender identities, one could study personal loans, but not compare fintech and bank-like lenders directly.

3.2 Definition of fintech lenders

Fintech lenders are defined as lenders that operate online, do not have a brick-and-mortar branch network, do not accept deposits, are not regulated by the Federal Reserve or the Office of the Comptroller of the Currency, and were founded after 2005. Large examples include LendingClub, SoFi, Avant, Upstart, Greensky, and similar online lenders.

Interpretation: This definition focuses on new nonbank online lenders. It excludes traditional banks that offer online channels but retain deposit-taking and branch-based business models.

3.3 Loan sample

The loan sample includes unsecured personal loans originated from 2012 to 2017. The paper restricts to loans above \$500, loans in the continental United States, loans with observed loan size and maturity, and fintech markets where the relevant lenders have meaningful origination volume. The final loan sample contains 3,792,757 loans to 1,882,286 borrowers. About 20.8% of loans are fintech loans.

Interpretation: Focusing on unsecured personal loans makes the comparison cleaner because collateral differences are not central. The main contract terms are loan size, maturity, and interest rate.

3.4 Loan and borrower variables

Key loan variables include:

- fintech loan indicator;
- loan size;
- maturity;
- note rate;
- loan delinquency indicator;
- origination year-month and geography.

Key borrower variables include:

- credit score;
- age of credit history;
- number of accounts;
- revolving utilization;
- prior delinquency;
- mortgage, student-loan, and transactor indicators;
- total, revolving, auto, student, and mortgage balances;
- age, gender, marital status, college status, income group, and occupation/professional status.

Interpretation: The data make it possible to compare fintech and non-fintech borrowers with similar credit scores and balances, but also to track what happens to those balances after origination.

3.5 Performance and dynamic outcomes

The main loan-performance outcome is whether the personal loan becomes delinquent within 15 months after origination. The borrower-level outcomes include delinquency on any account, credit score, total debt, auto loan balance, car-purchase proxy, and revolving debt.

The dynamic panel follows loans and borrowers from before origination through 15 months after origination:

$$t \in \{-3, -2, -1, 0, 1, \dots, 15\},$$

where $t = 0$ is loan origination.

Interpretation: The dynamic design distinguishes debt consolidation from subsequent over-borrowing. A loan can initially reduce revolving debt and improve the credit score but still lead to worse financial fragility later.

3.6 Matching samples

To address observable borrower differences, the paper uses several matched samples:

- propensity-score matching within zip codes;
- manual matching on zip code, age, and total indebtedness;
- entropy balancing, which reweights non-fintech observations to match fintech borrower characteristics.

Interpretation: Matching is used to make non-fintech borrowers look more like fintech borrowers before origination. The performance comparison then asks whether similarly observed borrowers perform differently after choosing different lender types.

4 Models and empirical specifications

4.1 Borrower-selection regressions

The paper first characterizes who borrows from fintech lenders by estimating regressions of the form:

$$100 \times Fintech_i = \alpha_{z(i), t(i)} + X_i' \beta + \varepsilon_i,$$

where $Fintech_i$ indicates that loan i is originated by a fintech lender, X_i contains borrower credit and demographic variables, and $\alpha_{z(i), t(i)}$ are zip-code-by-time fixed effects.

Interpretation: The coefficients describe which borrower traits are associated with fintech borrowing within the same local market and time period. They are descriptive, not causal treatment effects.

4.2 Loan-term regressions

Loan terms are compared using specifications such as:

$$Term_i = \alpha_{z(i),t(i)} + \beta Fintech_i + X_i' \Gamma + u_i,$$

where $Term_i$ is loan size, maturity, or note rate. Some specifications use borrower fixed effects for individuals who borrow from both fintech and non-fintech lenders.

Interpretation: Borrower fixed effects compare different loans taken by the same borrower. This is useful for asking whether fintech lenders provide different terms to the same individual.

4.3 Hard-information pricing and soft-information deficiency

To study screening, the paper compares how much of interest-rate variation is explained by observable credit-report variables:

$$Rate_i = \alpha_{z(i),t(i)} + f(X_i) + \varepsilon_i.$$

The key statistic is the R^2 from different specifications estimated separately for fintech and non-fintech loans.

Interpretation: A high R^2 for fintech rates means that hard information from the credit report explains much of fintech pricing. A lower R^2 for traditional lenders suggests that other information, possibly soft or relationship-based, matters more for their pricing.

4.4 Loan-performance regressions

The baseline performance model is:

$$100 \times LoanDLQ_i = \alpha_{g(i),t(i)} + \beta Fintech_i + X_i' \Gamma + L_i' \Lambda + \varepsilon_i,$$

where $LoanDLQ_i$ indicates delinquency within 15 months, X_i are borrower characteristics, L_i are loan terms, and fixed effects control for geography and origination time.

Interpretation: The coefficient β measures the difference in ex post delinquency between fintech and comparable non-fintech loans. The central finding is that $\beta > 0$ across several specifications.

4.5 Dynamic outcome regressions

For outcome Y_{it} in relative month t , the paper estimates:

$$Y_{it} = \alpha_i + \delta_t + \gamma_{m(t)} + \sum_{k=-3}^{15} \beta_k (Fintech_i \times \mathbf{1}\{m(t) = k\}) + \varepsilon_{it},$$

where α_i are loan fixed effects, δ_t are calendar-time fixed effects, and $\gamma_{m(t)}$ are relative-month fixed effects.

Interpretation: The dynamic coefficients show whether fintech borrowers diverge only after origination. Flat pre-period coefficients help support the matched comparison, while post-origination patterns reveal debt accumulation and default dynamics.

4.6 Pricing and ex post performance

The paper tests whether rates predict future default:

$$100 \times \text{LoanDLQ}_i = \alpha + \beta \text{Rate}_i + \theta \text{Fintech}_i + \rho(\text{Rate}_i \times \text{Fintech}_i) + X_i' \Gamma + \varepsilon_i.$$

Interpretation: If $\rho > 0$, fintech rates are more strongly linked to future default than non-fintech rates. This suggests that fintech lenders price risk on the intensive margin even if their borrowers default more often.

5 Identification and empirical design

5.1 What is identified

The paper does not estimate a randomized effect of receiving a fintech loan. Instead, it identifies conditional differences between fintech and non-fintech borrowers and loans using very rich credit-bureau controls, local-time fixed effects, matching, borrower fixed effects, and dynamic pre/post comparisons.

Interpretation: The main claim is empirical and diagnostic: fintech loans have worse ex post performance than observationally similar traditional loans, and the difference is not explained by standard observable credit characteristics or local credit-demand conditions.

5.2 Local-time comparisons

A central identifying device is the use of zip-code-by-time fixed effects:

$$\alpha_{z,t}.$$

These fixed effects compare borrowers in the same local credit market and same period, absorbing local economic shocks, lender entry timing, housing conditions, and time-varying demand for credit.

Interpretation: This reduces the concern that fintech loans perform worse simply because fintech lenders enter weaker local economies or periods of higher credit stress.

5.3 Rich borrower controls

The paper controls for credit scores, prior delinquencies, age of credit history, debt balances, number of accounts, utilization, demographics, loan size, maturity, and sometimes fixed effects for loan-size and maturity bins.

Interpretation: These controls approximate the hard information available to lenders. If fintech still predicts default after these controls, the remaining difference is consistent with unobserved risk, soft-information deficiency, or borrower selection into fintech credit.

5.4 Matched samples

The matching exercises strengthen the benchmark comparison. The propensity-score sample matches fintech borrowers to non-fintech borrowers in the same zip code using pre-origination credit and demographic variables. Manual matching and entropy balancing provide alternative ways to check that the fintech default premium does not depend on a single matching method.

Interpretation: The persistence of the default premium after matching is important because the raw fintech borrower pool differs from the traditional borrower pool. Matching makes the test closer to: among borrowers who look similar on the credit report, do fintech loans perform differently?

5.5 Borrower fixed effects

For borrowers who take multiple loans from different lender types, the paper includes borrower fixed effects:

$$100 \times \text{LoanDLQ}_{ij} = \alpha_i + \beta \text{Fintech}_{ij} + X'_{ij}\Gamma + \varepsilon_{ij}.$$

Interpretation: Borrower fixed effects absorb time-invariant borrower traits. If the fintech coefficient remains positive, the evidence is harder to explain by permanent borrower quality alone, although time-varying shocks can still matter.

5.6 Dynamic evidence and pre-trends

The dynamic figures compare fintech and matched non-fintech borrowers before and after origination. The most useful visual check is whether the groups look similar before loan origination and then diverge afterward.

Interpretation: The dynamic patterns show that fintech borrowers' credit scores initially improve and debts initially fall, consistent with debt consolidation. The later reversal is important: it suggests the fintech loan relaxes a credit constraint that allows additional consumption and debt accumulation.

5.7 Threats to identification

- **Time-varying unobservables.** Borrowers may turn to fintech lenders when they experience unobserved financial stress not captured by the credit report.
- **Unobserved fees.** The paper observes note rates but not all origination fees or other charges, so the full price of fintech credit may be higher than the measured rate.
- **Selection into lender type.** The decision to borrow from fintech is not random; borrower preferences, urgency, and financial sophistication may differ.
- **General equilibrium.** Fintech entry may change traditional lender behavior, so comparisons are conditional on the market structure during 2012–2017.
- **Soft information.** The paper cannot directly observe the soft information used by traditional lenders. It infers soft-information differences from the relative explanatory power of hard credit variables and from ex post performance.

Interpretation: The strongest evidence is the combination of local-time fixed effects, matched samples, borrower fixed effects, and post-origination dynamics. The paper is careful to interpret the results as evidence on selection, screening, pricing, and borrower behavior, not as a randomized causal estimate of fintech lending.

6 Tables and figures

6.1 Table 1: Summary Statistics; Figure 1: Fintech loan market

Read:

- Table 1 shows that the sample is large: 3.79 million loans, with fintech loans representing 20.8% of the loan-level sample.

- The average loan size is about \$8,492 and the average maturity is roughly 39 months.
- The average note rate is about 13.1%, and the average loan-delinquency indicator is 1.45%.
- Figure 1 shows fintech lenders expanding rapidly in loan counts and market share between 2012 and 2017.

Detailed interpretation: Table 1 establishes the scale and breadth of the sample. Figure 1 gives the market backdrop: fintech lending is not a tiny niche in this data; it is a rapidly growing segment of unsecured personal credit.

Economic message: The paper studies a market whose composition is changing quickly. That growth makes the screening and performance questions economically important.

Table 1

Summary Statistics

Statistic	N	Mean	St. Dev.	Min	Pctl(25)	Median	Pctl(75)	Max
Fintech loan indicator	3,792,757	0.208	0.406	0	0	0	0	1
Loan size	3,792,757	\$8,492	\$11,822	\$1	\$1,575	\$5,446	\$11,869	\$5,850,000
Maturity	3,792,757	39.185	26.26	1	13	36	60	999
Note rate	2,218,740	13.108	9.818	0	7.033	11.55	18.818	49.996
Borrower DLQ indicator	3,792,757	21.495	41.079	0	0	0	0	100
Loan DLQ indicator	3,792,757	1.452	11.964	0	0	0	0	100
No. of accounts	3,709,883	22.878	13.12	0	13	21	30	91
Total balance	3,709,883	\$106,986	\$146,034	\$-	\$13,490	\$45,595	\$157,011	\$8,857,760
Auto balance	3,709,883	\$11,739	\$16,168	\$-	\$-	\$5,731	\$18,650	\$1,997,020
Revolving balance	3,709,883	\$9,749	\$24,402	\$-	\$456	\$3,550	\$10,754	\$6,089,626
Student balance	3,709,883	\$8,061	\$25,897	\$-	\$-	\$-	\$-	\$980,313
Mortgage balance	3,709,883	\$73,061	\$128,126	\$-	\$-	\$-	\$117,168	\$8,175,626
Student indicator	3,792,757	0.281	0.449	0	0	0	1	1
Mortgage indicator	3,792,757	0.532	0.499	0	0	1	1	1
Credit score	3,791,076	653.836	97.873	0	596	658	722	840
Inquiries	3,709,883	1.253	1.611	0	0	1	2	74
Age of credit	3,709,883	188.407	100.958	0	122	170	240	1050
Revolving utilization	3,672,635	0.434	0.344	0	0.079	0.419	0.74	1.111
Age	3,637,150	48.785	14.323	14	38	48	59	99
Young indicator	3,792,757	0.412	0.492	0	0	0	1	1
Male indicator	3,792,757	0.515	0.5	0	0	1	1	1
Married indicator	3,792,757	0.573	0.495	0	0	1	1	1
College indicator	3,792,757	0.238	0.426	0	0	0	0	1
Professional indicator	3,792,757	0.084	0.277	0	0	0	0	1
High Income indicator	3,792,757	0.209	0.406	0	0	0	0	1
Prior DLQ indicator	3,709,883	0.168	0.374	0	0	0	0	1
Transactor indicator	3,782,705	0.067	0.251	0	0	0	0	1

This table reports summary statistics of loan-level data. Our loan sample is a random sample of all unsecured personal loans originated by fintech and by non-fintech lenders from 2012–2017. The (original) note rate is calculated using scheduled payment, loan term and loan size. *Borrower DLQ indicator* is an indicator for borrowers who have any positive delinquent balance on their credit report. *High income indicator* is an indicator for households whose income is more than \$100,000. *Professional indicator* is an indicator for households whose heads work in the professional, technical, and management occupations. *Fintech loan indicator* is an indicator

(a) Table 1: Summary Statistics. The table reports loan, borrower, debt-balance, credit-history, and demographic variables for the full loan sample. It shows the baseline scale of fintech personal lending and the borrower characteristics available for matching and controls.

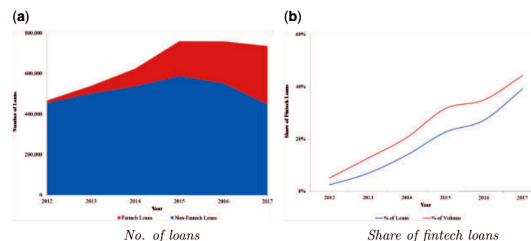


Figure 1

Fintech loan market

This figure plots the number of loans originated by fintech lenders and non-fintech lenders based on our loan sample. The largest fintech lenders include LendingClub Corporation, Greensky Financial, SOFI Lending

(b) Figure 1: Fintech loan market. The figure plots fintech and non-fintech originations over time and shows the increase in fintech market share.

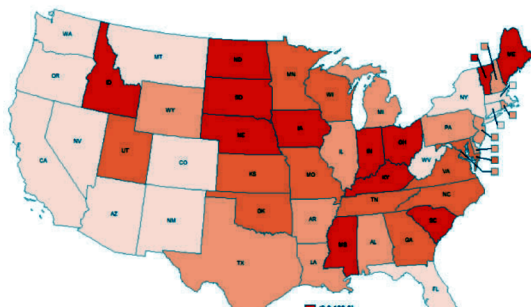
6.2 Figure 2: Fintech volume growth from 2013 to 2017; Figure 3: Evolution of borrower profile after market entry

Read:

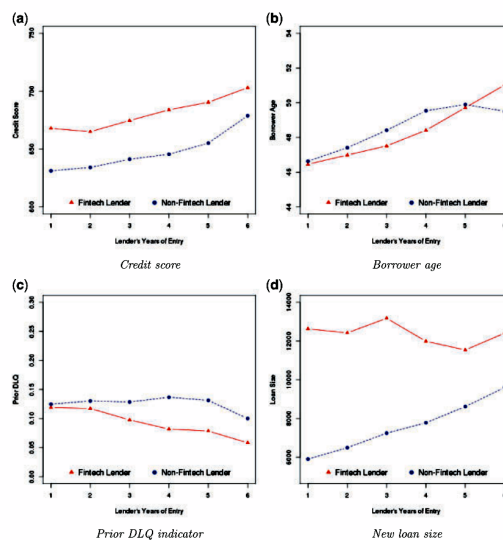
- Figure 2 maps geographic growth in fintech origination volume. Growth is broad but not uniform across states.
- Figure 3 shows that fintech lenders initially serve borrowers with weaker profiles and later move toward safer borrowers.
- Credit scores rise with lender experience, prior delinquency falls, and borrower age increases over the entry cycle.
- Loan sizes for fintech lenders remain more standardized than for non-fintech lenders, consistent with more algorithmic lending menus.

Detailed interpretation: The market-entry evidence is central to the “lax screening versus cream-skimming” question. Fintech lenders do not simply begin by targeting the safest borrowers; they seem to start with riskier or more marginal borrowers and then move up the credit-quality distribution.

Economic message: The fintech strategy looks like dynamic market penetration: build data, brand, and scale first, then expand toward better borrowers.



(a) Figure 2: Fintech volume growth from 2013 to 2017. The map shows state-level differences in fintech loan growth and highlights the geographic diffusion of online lending.



(b) Figure 3: Evolution of borrower profile after market entry. The panels compare credit scores, age, prior delinquency, and loan sizes as lenders gain experience in a state.

6.3 Table 2: Evolution of fintech loan attributes; Figure 4: Evolution of credit score distribution

Read:

- Table 2 reports how fintech loan and borrower attributes change with the number of years a fintech lender has operated in a state.
- Fintech borrowers become older and safer over time; the share of younger borrowers and the probability of prior delinquency decline.
- Figure 4 shows that fintech lending is concentrated in the middle of the credit-score distribution, especially around the 640–720 range.
- The distributional evidence weakens a pure cream-skimming interpretation because fintech does not only target the very top credit-score borrowers.

Detailed interpretation: The table and figure jointly show that fintech lenders serve a broad but specific borrower segment: borrowers with access to credit and mid-range scores, not purely excluded borrowers and not only top-quality prime borrowers.

Economic message: Fintech’s comparative advantage may be speed, convenience, and standardized underwriting rather than discovering invisible prime borrowers.

Table 2
Evolution of fintech loan attributes

	Fintech lender's years of entry in each state					
	1	2	3	4	5	6
Age	46.46	46.988	47.517	48.416	49.717	51.023
% Age ≤ 45	0.488	0.473	0.456	0.429	0.388	0.354
% Professional	0.111	0.112	0.114	0.114	0.116	0.119
Credit score	667.9	665.0	674.7	683.9	690.4	703.0
% Credit score ≤ 620	0.3	0.3	0.2	0.2	0.2	0.1
Age of credit	196.6	199.1	204.8	213.6	220.1	230.2
No. of accounts	23.6	23.0	23.3	23.0	22.7	23.3
Total balance	\$146,406	\$144,227	\$144,776	\$141,573	\$143,742	\$158,996
Rev. utilization	0.498	0.516	0.484	0.467	0.442	0.409
No of inquiries	1.212	1.037	0.947	0.792	0.712	0.625
Prior DLQ	0.12	0.12	0.10	0.08	0.08	0.06
Loan size	\$12,633	\$12,424	\$13,177	\$11,991	\$11,542	\$12,411
Maturity	47.0	47.7	52.2	53.0	56.0	59.6
Note rate	15.35	17.41	17.50	15.67	15.93	15.25

(a) Table 2: Evolution of fintech loan attributes. The table summarizes borrower and loan attributes by fintech lender years of entry in each state.

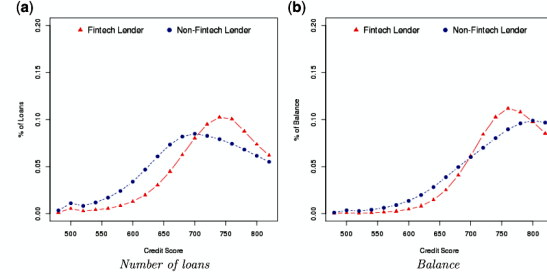


Figure 4
Evolution of credit score distribution

(b) Figure 4: Evolution of credit score distribution. The figure compares fintech and non-fintech loan density by credit score for counts and balances.

6.4 Table 3: Individual characteristics of fintech borrowers; Table 4: Loan terms

Read:

- Table 3 shows that fintech borrowers are associated with higher credit scores, longer credit histories, more accounts, higher revolving utilization, and lower prior delinquency after zip-by-time controls.
- Demographically, fintech borrowers are more likely to be college educated, high income, married, young, and professional.
- Table 4 shows that fintech loans are larger and have longer maturities in broad comparisons.
- For low-score borrowers, fintech rates are higher; for high-score borrowers, fintech rates can be lower.
- The same-borrower sample reduces some differences, showing that part of the raw pricing difference reflects borrower selection.

Detailed interpretation: These tables show that fintech borrowers are not the most excluded households. They often have extensive credit histories and substantial credit access. However, pricing patterns show that fintech lenders charge a premium where borrower risk is harder to assess, especially for thin-file and low-score borrowers.

Economic message: The results suggest a technology that prices hard information aggressively but may not replace relationship or soft-information screening.

Table 3
Individual characteristics of fintech borrowers

A. Credit attributes

Dep. var.	$I(Fintech) \times 100$								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Credit score	0.019*** (0.001)								0.007*** (0.001)
Age of credit history		0.020*** (0.000)							0.014*** (0.000)
No. of accounts			0.011*** (0.000)						0.004*** (0.000)
Rev. utilization				0.006*** (0.000)					0.008*** (0.000)
Prior DLQ indicator					-0.045*** (0.001)				-0.036*** (0.001)
Transactor indicator						0.039*** (0.001)			0.030*** (0.001)
Student loan indicator							0.043*** (0.001)		0.046*** (0.001)
Mortgage indicator								0.041*** (0.001)	0.024*** (0.001)
Zip $\times$ Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	3,791,076	3,709,883	3,709,883	3,672,635	3,709,883	3,782,705	3,792,758	3,792,758	3,663,667
R^2	0.382	0.383	0.382	0.381	0.382	0.381	0.382	0.382	0.387

B. Demographics

Dep. var.	$I(Fintech) \times 100$						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Male indicator	0.004*** (0.001)						0.001*** (0.001)
Married indicator		0.016*** (0.001)					0.009*** (0.001)
College indicator			0.042*** (0.001)				0.031*** (0.001)
High income indicator				0.046*** (0.001)			0.038*** (0.001)
Young indicator					0.010*** (0.001)		0.012*** (0.001)
Professional indicator						0.045*** (0.001)	0.031*** (0.001)
Zip $\times$ Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	3,792,757	3,792,757	3,792,757	3,792,757	3,792,757	3,792,757	3,792,757
R^2	0.38	0.38	0.381	0.382	0.38	0.381	0.383

(a) Table 3: Individual characteristics of fintech borrowers. The table regresses the fintech-loan indicator on credit attributes and demographics, controlling for local time effects.

Table 4
Loan terms

A. Overall sample

Sample	Borrowers that have fintech or bank loans			Borrowers that have fintech and bank loans		
Dep. Var.	Loan size	Loan term	Note rate	Loan size	Loan term	Note rate
	(1)	(2)	(3)	(4)	(5)	(6)
Fintech loan indicator	1,562.587*** (30.610)	10,506*** (0.096)	1,498*** (0.062)	2,939.567*** (38.299)	-1,176*** (0.082)	-0.135** (0.054)
Loan size			-1.338*** (0.024)			-0.103*** (0.020)
Maturity			0.082*** (0.002)			0.051*** (0.003)
FICO	3,129.673*** (16.015)	8,658*** (0.040)	-2,815*** (0.028)	15,639*** (0.922)	0.040*** (0.002)	-0.036*** (0.002)
Age of credit	1,031.716*** (9.682)	1,970*** (0.021)	0.019 (0.013)	1,036*** (0.344)	0.002 (0.001)	-0.001* (0.001)
Age	-739.777*** (7.896)	-0.378*** (0.022)	0.088*** (0.013)			
Time FE	No	No	No	Yes	Yes	Yes
Zip $\times$ Time FE	Yes	Yes	Yes	No	No	No
Individual FE	No	No	No	Yes	Yes	Yes
N	3,559,985	3,559,985	2,079,676	253,695	253,695	163,436
R^2	0.371	0.526	0.565	0.704	0.576	0.765

(Continued)

(b) Table 4: Loan terms, Panel A. The table compares loan size, maturity, and note rates for fintech and non-fintech loans in the overall and same-borrower samples.

Table 4
Continued

B. By borrower attributes

Dep. var.	Note rate							
Sample	Credit score		Age		County fintech share		Thin credit	
	High	Low	Old	Young	High	Low	No	Yes
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Fintech loan indicator	-1.563*** (0.040)	2.794*** (0.103)	2.506*** (0.130)	1.153*** (0.055)	1.425*** (0.085)	0.719*** (0.091)	1.404*** (0.061)	5.436*** (1.373)
Loan size	-0.035** (0.015)	-2.828*** (0.044)	-2.465*** (0.049)	-0.950*** (0.029)	-0.732*** (0.019)	-1.646*** (0.050)	-1.265*** (0.023)	-5.216*** (0.828)
Maturity	0.003*** (0.001)	0.155*** (0.003)	0.171*** (0.004)	0.026*** (0.002)	0.020*** (0.003)	0.119*** (0.003)	0.078*** (0.002)	0.236*** (0.025)
Credit score	-4.192*** (0.036)	-1.061*** (0.034)	-2.178*** (0.038)	-3.659*** (0.034)	-5.089*** (0.057)	-2.017*** (0.029)	-2.896*** (0.028)	-0.106 (0.172)
Age of credit	-0.052*** (0.015)	0.220*** (0.019)	0.055*** (0.021)	-0.120*** (0.017)	-0.239*** (0.021)	0.132*** (0.015)	-0.032*** (0.013)	0.74 (0.740)
Age	0.217*** (0.017)	-0.018 (0.016)	0.057*** (0.018)	0.320*** (0.018)	0.370*** (0.021)	-0.025* (0.015)	0.111*** (0.013)	-0.304*** (0.091)
Zip $\times$ Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	730,390	1,349,286	931,309	1,148,367	704,991	1,343,393	1,988,624	91,052
R^2	0.653	0.674	0.718	0.617	0.525	0.594	0.564	0.939

The table reports the regression results examining the difference in the loan terms between the fintech and non-fintech loans in our sample. The dependent variable in panel A is reported in the column title: loan size, loan term, and note rate. The dependent variable in panel B is note rate of the loan. *Note rate* is calculated using scheduled payment, loan term, and loan size. In panel B, the regressions are based on split samples by borrower attributes (credit score, age, county fintech share and age of credit). *High credit score* is defined as the borrowers with a high score above 700. *Young* is defined as borrowers younger than 45 at the time of loan origination. *County* is defined to have high fintech share if the fintech share in previous three months is above 20%. Borrowers are considered to have thin credit file if their credit history is shorter than 48 months. In addition to the fintech loan indicator, we also control for borrowers' credit score, age of credit history, and age. For the regression of note

Table 4: Loan terms, Panel B. This continuation splits note-rate comparisons by credit score, age, county fintech share, and thin credit file. The key reading is that fintech pricing is more favorable for high-score borrowers and more expensive for riskier or harder-to-evaluate borrowers.

6.5 Table 5: Soft information: R^2 of different specifications explaining interest rates

Read:

- Table 5 compares how much of rate variation is explained by observable credit-report variables.
- For fintech lenders, hard-information specifications explain a large share of the variation in interest rates.
- For non-fintech lenders, observable hard information explains less of rate variation.
- The difference is consistent with fintech lenders relying more on standardized hard data and traditional lenders using more non-credit-report information.

Detailed interpretation: This is one of the most important mechanism tables. It does not directly observe soft information, but it infers its role by asking whether hard information is enough to explain pricing. If traditional rates are less explained by hard variables, traditional lenders may be using relationships, branch knowledge, or other borrower information not captured in the bureau file.

Economic message: Fintech may improve speed and scalability by standardizing underwriting, but standardization can miss borrower traits that matter for repayment.

Table 5
Soft information: R^2 of different specifications explaining interest rates

A. R^2 based on overall sample

				R^2			
Specification				Sample			Fintech -
Controls	Time FE	Zip-Time FE	Lender	All	Fintech	Non-fintech	Non-fintech
Credit score	Yes	No	No	0.033	0.437	0.014	0.423***
Credit score, size	Yes	No	No	0.034	0.445	0.019	0.426***
Credit score, size	Yes	Yes	No	0.533	0.768	0.576	0.192***
All	Yes	No	No	0.211	0.49	0.281	0.209***
All	Yes	Yes	No	0.580	0.79	0.645	0.145***
Nonlinear	No	Yes	No	0.256	0.505	0.324	0.181***
Nonlinear	No	Yes	No	0.597	0.796	0.597	0.199***
Nonlinear	No	Yes	Yes	0.833	0.855	0.861	-0.006

B. R^2 by years of entry

Specification		Sample			Fintech -
Controls	Zip-YM FE	Years of Entry	Fintech	Non-fintech	Non-fintech
Nonlinear	Yes	1	0.975	0.748	0.227***
Nonlinear	Yes	2	0.980	0.741	0.239***
Nonlinear	Yes	3	0.969	0.726	0.243***
Nonlinear	Yes	4	0.918	0.686	0.232***
Nonlinear	Yes	5	0.868	0.591	0.277***
Nonlinear	Yes	6	0.787	0.533	0.254***

This table shows the R^2 for different specifications of a regression similar to Table 4. Data includes all loans in our loan-level sample. Panel A shows pooled regressions based on entire sample, non-fintech, and fintech subsamples, respectively. In addition to credit score and loan size, we also control for loan maturity, length of credit history, number of accounts, revolving utilization, prior DLQ indicator, transactor indicator, mortgage indicator, student loan indicator, and age in the “All” specification. Nonlinear regressions include nonlinear splines of credit score and loan size. Tests of significance of R^2 differences follow Erickson and Whited (2002).

Table 5: Soft information. The table reports R^2 values from note-rate regressions with different sets of hard-information variables and fixed effects. Higher explanatory power for fintech pricing indicates stronger reliance on observed credit-report variables.

6.6 Table 6: Loan performance based on entire loan sample; Table 7: Loan performance based on matched samples

Read:

- Table 6 shows that fintech loans are more likely to become delinquent in the full sample, even after controls and fixed effects.
- The delinquency premium is particularly large for low-score borrowers and thin-file borrowers.
- Table 7 shows that the result survives propensity-score matching, manual matching, and entropy balancing.
- The matched-sample estimate is economically large because the sample mean delinquency rate is low.

Detailed interpretation: The tables are the core evidence against a pure cream-skimming story. If fintech lenders were simply selecting better borrowers, matched fintech loans should perform at least as well as comparable non-fintech loans. Instead, similar fintech borrowers have higher subsequent delinquency.

Economic message: The evidence is consistent with adverse selection or soft-information deficiency: fintech lenders can price hard risk, but borrowers who choose fintech may still carry unobserved repayment risk.

Table 6
Loan performance based on entire loan sample

A. Overall sample

Dep. var.	$I(LoanDLQ_t) \times 100$				
	(1)	(2)	(3)	(4)	(5)
Fintech loan indicator	0.845*** (0.027)	1.132*** (0.029)	1.157*** (0.029)	1.212*** (0.037)	1.209*** (0.037)
Loan size		-0.105*** (0.017)	-0.095*** (0.016)	-0.098*** (0.018)	
Maturity		0.001 (0.000)	-0.001* (0.000)	-0.001 (0.001)	
Credit score		-1.336*** (0.014)	-1.354*** (0.014)	-1.350*** (0.017)	-1.401*** (0.017)
Age of credit		-0.033*** (0.008)	-0.041*** (0.008)	-0.051*** (0.010)	-0.066*** (0.010)
No. of accounts		-0.278*** (0.007)	-0.276*** (0.007)	-0.276*** (0.009)	-0.223*** (0.009)
Age		-0.080*** (0.008)	-0.071*** (0.008)	-0.078*** (0.010)	-0.058*** (0.010)
Time FE	Yes	Yes	No	No	No
County FE	Yes	Yes	No	No	No
County $\times$ Time FE	No	No	Yes	No	No
Zip $\times$ Time FE	No	No	No	Yes	Yes
Loan size bin FE	No	No	No	No	Yes
Maturity bin FE	No	No	No	No	Yes
N	3,792,757	3,559,985	3,559,985	3,559,985	3,559,985
R ²	0.007	0.017	0.041	0.235	0.237

(Continued)

show that fintech loans severely underperform non-fintech loans for similar borrowers.¹⁴

(a) Table 6: Loan performance based on entire loan sample, Panel A. The table estimates delinquency differences between fintech and non-fintech loans under increasingly rich controls.

Table 6
Continued

B. By borrower attributes

Dep. var.	$I(LoanDLQ_t) \times 100$							
	Credit score		Main lender		County fintech share		Thin credit	
	High	Low	No	Yes	High	Low	No	Yes
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Fintech loan indicator	0.029 (0.026)	2.186*** (0.068)	1.109*** (0.069)	1.103*** (0.047)	0.828*** (0.057)	1.497*** (0.049)	1.138*** (0.036)	5.847*** (1.593)
Loan size	0.005 (0.017)	-0.052*** (0.056)	-0.116*** (0.042)	-0.090*** (0.024)	-0.054* (0.028)	-0.109*** (0.015)	-0.093*** (0.018)	-0.654 (0.703)
Maturity	-0.001** (0.001)	0.011*** (0.001)	0.005*** (0.001)	-0.006*** (0.001)	0.003*** (0.001)	-0.002*** (0.001)	-0.001 (0.001)	0.015 (0.016)
Credit score	-0.527*** (0.029)	-1.548*** (0.029)	-0.997*** (0.024)	-1.799*** (0.029)	-2.147*** (0.039)	-1.063*** (0.017)	-1.329*** (0.017)	-1.340*** (0.265)
Age of credit	-0.004 (0.013)	-0.183*** (0.017)	-0.095*** (0.016)	-0.053*** (0.017)	-0.070*** (0.022)	-0.036*** (0.011)	-0.018* (0.010)	-1.476 (1.347)
No. of accounts	-0.066*** (0.013)	-0.285*** (0.012)	-0.151*** (0.012)	-0.280*** (0.018)	-0.294*** (0.023)	-0.255*** (0.010)	-0.257*** (0.009)	-0.17 (0.243)
Age	0.065*** (0.014)	-0.126*** (0.015)	-0.081*** (0.017)	-0.009 (0.017)	-0.005 (0.023)	-0.108*** (0.011)	-0.072*** (0.01)	-0.273** (0.139)
Zip $\times$ Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	1,183,605	2,376,380	1,613,759	1,946,226	1,076,085	2,431,632	3,407,831	152,154
R ²	0.429	0.313	0.353	0.347	0.256	0.22	0.241	0.815

The table reports the regression results examining the difference in the ex post performance between the fintech and non-fintech loans in our sample. The dependent variable in all regressions is an indicator whether the loan has ever become delinquent from origination through 15 months after the origination. Regressions in panel A are based on the entire sample, and those in panel B are based on split samples by borrower attributes (credit score, main lender, county fintech share and thin credit). *High credit score* is defined as borrowers with a high score, above 700. *Young* is defined as borrowers younger than 45 at the time of loan origination. *County* is defined as having high fintech share if the fintech share in previous three months is above 20%. Borrowers are considered to have a thin credit file if their length of credit history is less than 48 months. In addition to the fintech loan indicator, we also control for loan size and loan maturity. *Borrowers' credit score* is the score of credit history, measured

(b) Table 6: Loan performance based on entire loan sample, Panel B. The continuation shows heterogeneity by score, main lender status, local fintech share, and thin credit.

Table 7
Loan performance based on matched samples

A. PSM-matched sample

Dep. var.	$I(\text{LoanDLQ}_i) \times 100$				
	(1)	(2)	(3)	(4)	(5)
Fintech loan indicator	1.739*** (0.061)	0.690*** (0.059)	0.576*** (0.065)	0.558*** (0.142)	1.102*** (0.149)
Loan size		-0.339*** (0.085)	-0.293*** (0.084)	-0.244 (0.196)	
Maturity		0.009*** (0.001)	0.011*** (0.001)	0.014*** (0.003)	
Credit score		-3.723*** (0.063)	-3.784*** (0.065)	-4.144*** (0.134)	-3.832*** (0.117)
Age of credit		-0.212*** (0.029)	-0.203*** (0.031)	-0.285*** (0.071)	-0.261*** (0.070)
No. of accounts		-0.444*** (0.040)	-0.500*** (0.042)	-0.540*** (0.090)	-0.412*** (0.083)
Age		0.135*** (0.034)	0.145*** (0.036)	0.196** (0.080)	0.163** (0.080)
Time FE	Yes	Yes	No	Yes	No
County FE	Yes	Yes	No	No	No
County $\times$ Time FE	No	No	Yes	No	No
Zip $\times$ Time FE	No	No	No	Yes	Yes
Loan Size Bin FE	No	No	No	No	Yes
Maturity Bin FE	No	No	No	No	Yes
N	418,626	418,626	418,626	418,626	418,626
R^2	0.012	0.048	0.134	0.55	0.553

(Continued)

Table 7
(Continued)

B. Two alternative samples

Dep. var.	$I(\text{LoanDLQ}_i) \times 100$							
	Manually matched				Entropy balancing			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Fintech loan indicator	1.583*** (0.054)	1.548*** (0.057)	1.645*** (0.096)	1.713*** (0.104)	2.241*** (0.037)	2.215*** (0.038)	2.358*** (0.051)	2.422*** (0.052)
Loan size		-0.415*** (0.026)	-0.365*** (0.027)	-0.311*** (0.044)	-0.041** (0.018)	-0.049** (0.020)	-0.083*** (0.026)	
Maturity		0.008*** (0.001)	0.007*** (0.001)	0.007*** (0.002)	-0.001** (0.001)	-0.001** (0.001)	-0.001 (0.001)	
Credit score		-2.376*** (0.039)	-2.409*** (0.041)	-2.402*** (0.061)	-2.373*** (0.061)	-1.851*** (0.022)	-1.859*** (0.023)	-1.766*** (0.027)
Age of credit		-0.203*** (0.025)	-0.188*** (0.027)	-0.209*** (0.047)	-0.196*** (0.047)	-0.159*** (0.013)	-0.157*** (0.014)	-0.144*** (0.018)
No. of accounts		-0.287*** (0.025)	-0.322*** (0.026)	-0.300*** (0.044)	-0.273*** (0.044)	-0.221*** (0.013)	-0.231*** (0.013)	-0.232*** (0.017)
Age		-0.001 (0.027)	0.001 (0.028)	-0.008 (0.050)	0.025 (0.050)	0.028** (0.014)	0.027* (0.014)	-0.016 (0.018)
Time FE	Yes	No	No	No	Yes	No	No	No
County FE	Yes	No	No	No	Yes	No	No	No
County $\times$ Time FE	No	Yes	No	No	No	Yes	No	No
Zip $\times$ Time FE	No	No	Yes	Yes	No	No	Yes	Yes
Loan size bin FE	No	No	No	Yes	No	No	No	Yes
Maturity bin FE	No	No	No	Yes	No	No	No	Yes
N	681,274	681,274	681,274	681,274	2,937,580	2,937,580	2,937,580	2,937,580
R^2	0.028	0.1	0.438	0.44	0.025	0.068	0.354	0.356

(Continued)

(a) Table 7: Loan performance based on matched samples, Panel A. The PSM-matched sample compares fintech loans with observationally similar non-fintech loans.

(b) Table 7: Loan performance based on matched samples, Panel B. Manual matching and entropy balancing confirm the higher fintech delinquency result.

6.7 Table 8: Dynamic effect; Figure 5: Dynamics of loan and borrower performance; Figure 6: Dynamics of borrower consumption

Read:

- Table 8 and Figure 5 show that fintech borrowers initially improve credit scores and reduce debt, but these gains reverse after a few months.
- Loan delinquency and borrower-level delinquency rise after origination for fintech borrowers relative to matched non-fintech borrowers.
- Total debt and revolving debt decline initially, consistent with debt consolidation, then rise again.
- Figure 6 shows greater durable-consumption behavior, including increased car-purchase probability.

Detailed interpretation: The dynamic results explain why fintech borrowers default more. The loans may initially be used to pay down expensive revolving debt, but easier liquidity also supports additional consumption. The pattern is therefore not simply bad underwriting at origination; it is also post-origination borrower behavior.

Economic message: Fintech credit can relax borrowing constraints in a way that raises short-run liquidity and longer-run fragility.

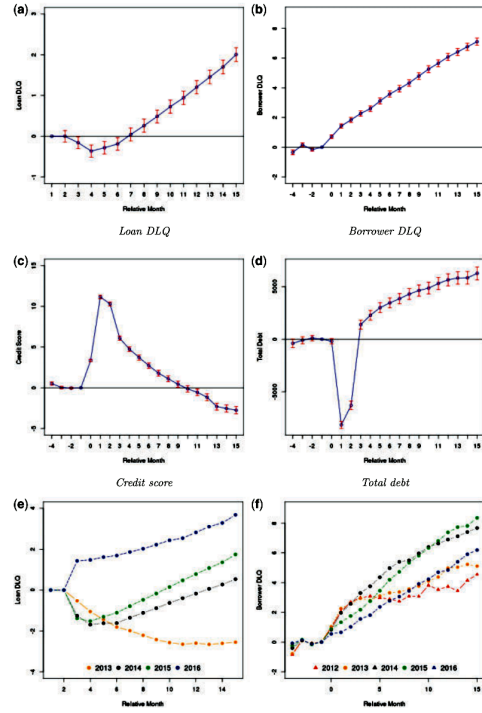
Table 8
Dynamic effect

Dep. var.	Loan DLQ	Borrower DLQ	Credit score	Total debt	New car	Revolving debt
	(1)	(2)	(3)	(4)	(5)	(6)
Fintech $\times$ Rel Qtr = -1	-0.148** (0.061)	0.051 (0.068)	109.927 (93.669)	-0.022 (0.048)	-7.941 (13.368)	
Fintech $\times$ Rel Qtr = 1	2.042*** (0.060)	7.438*** (0.124)	-3,212.081*** (163.740)	0.061 (0.045)	-669.683*** (33.921)	
Fintech $\times$ Rel Qtr = 2	-0.241*** (0.031)	3.288*** (0.082)	3.445*** (0.171)	3,007.067*** (227.405)	0.250*** (0.044)	-163.694*** (46.62)
Fintech $\times$ Rel Qtr = 3	0.297*** (0.042)	4.543*** (0.093)	0.833*** (0.197)	4,347.042*** (248.564)	0.101** (0.044)	202.527*** (54.295)
Fintech $\times$ Rel Qtr = 4	0.987*** (0.050)	5.849*** (0.100)	-0.894*** (0.214)	5,355.385*** (274.700)	0.039 (0.045)	367.762*** (62.229)
Fintech $\times$ Rel Qtr = 5	1.746*** (0.055)	6.946*** (0.107)	-2.808*** (0.232)	6,058.412*** (295.757)	0.032 (0.044)	465.642*** (70.935)
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Relative Qtr FE	Yes	Yes	Yes	Yes	Yes	Yes
N	5,457,382	8,374,729	8,374,729	8,354,735	7,941,055	8,354,735
R ²	0.603	0.608	0.879	0.924	0.062	0.916

The table reports the difference in the dynamics of several credit attributes between the fintech and non-fintech loans in our sample. The regression is based on the loan-month panel from 3 months before through 15 months after the origination of the loan, restricted to the loans in the PSM-matched sample. The dependent variable is the column title. Our main explanatory variable is the fintech loan indicator interacted with the relative quarter dummies. In all regressions, we control for individual, calendar time, and relative quarter fixed effects. Standard errors are double clustered at county and origination year-month levels. * $p < .1$; ** $p < .05$; *** $p < .01$

third quarter. Furthermore, their underperformance becomes more pronounced

(a) Table 8: Dynamic effect. The table estimates event-time differences in delinquency, credit score, debt balances, car purchases, and revolving debt for matched fintech and non-fintech borrowers.



(b) Figure 5: Dynamics of loan and borrower performance. The figure plots month-by-month fintech interaction coefficients for loan performance, borrower performance, credit score, debt, and cohort-specific delinquency.

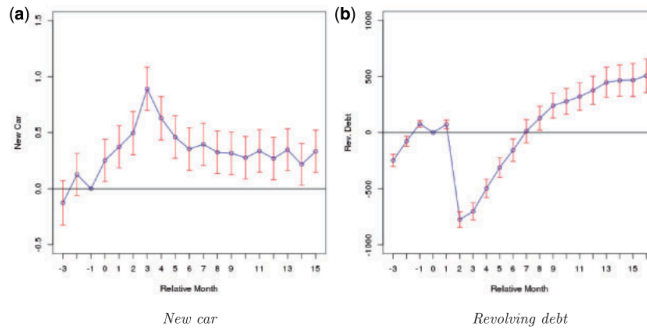


Figure 6

Dynamics of borrower consumption

This figure plots the coefficient on the interaction of fintech loan indicator and relative month dummies from the regression examining the difference in the spending dynamics between the fintech and non-fintech borrowers in our sample. The regression is based on the loan-month panel from 3 months before through 15 months after the origination of the loan, restricted to the loans in the PSM-matched sample. The dependent variable for each chart is the figure title. We control for individual loan, calendar time, and relative month fixed effects. Standard errors are double clustered at county and origination year-month levels.

use the additional funds to repay their credit cards, but then start financing

Figure 6: Dynamics of borrower consumption. The panels show that fintech borrowers have greater post-origination car-purchase probability and revolving-debt changes, indicating that fintech loans may finance consumption as well as consolidation.

6.8 Table 9: Pricing and ex post performance; Table 10: Repeat borrowers; Table 11: Job loss and fintech loans

Read:

- Table 9 shows that note rates predict future delinquency, and the rate-default link is stronger for fintech loans.
- Table 10 shows strong persistence in fintech borrowing: borrowers with a prior fintech loan are much more likely to borrow from fintech again.
- Table 11 studies job loss and shows that borrowers facing adverse labor-market shocks are more likely to use fintech credit.
- Together, these tables show that high default risk may be priced and that fintech lenders may gain repeat-customer value.

Detailed interpretation: Table 9 is important because it prevents the interpretation that fintech lenders are simply making unpriced mistakes. Fintech borrowers default more, but fintech rates also appear to forecast default better. Table 10 adds that lender relationships may persist even without physical branches. Table 11 connects fintech usage to liquidity shocks and borrower distress.

Economic message: The fintech business model can survive higher defaults if risks are priced and repeat-borrower value is high. The borrower welfare implications are more ambiguous because easier access can help after shocks but also increases fragility.

Table 9
Pricing and ex post performance

Dep. var.	$I(LoanDLQ_t) \times 100$					
	(1)	(2)	(3)	(4)	(5)	(6)
Note rate	0.058*** (0.001)	0.054*** (0.001)	0.054*** (0.001)	0.051*** (0.001)	0.051*** (0.002)	0.048*** (0.002)
× Fintech loan indicator		0.020*** (0.002)		0.012*** (0.002)		0.010*** (0.003)
Loan size			0.018* (0.009)	0.009 (0.009)	0.018 (0.016)	0.012 (0.015)
Maturity			-0.008*** (0.001)	-0.008*** (0.001)	-0.008*** (0.001)	-0.008*** (0.001)
Credit score			-0.846*** (0.015)	-0.834*** (0.015)	-0.961*** (0.023)	-0.952*** (0.023)
Age of credit			-0.029*** (0.009)	-0.029*** (0.009)	-0.054*** (0.014)	-0.054*** (0.014)
No. of accounts			-0.144*** (0.009)	-0.146*** (0.009)	-0.143*** (0.012)	-0.144*** (0.012)
Age			-0.044*** (0.009)	-0.042*** (0.009)	-0.030** (0.014)	-0.028** (0.014)
Time FE	Yes	Yes	Yes	Yes	No	No
Zip time FE	No	No	No	No	Yes	Yes
N	2,218,740	2,218,740	2,079,676	2,079,676	2,079,676	2,079,676
R ²	0.008	0.008	0.015	0.015	0.314	0.314

The table reports the regression results examining the difference in the relationship between ex post performance and pricing between the fintech and non-fintech personal loans in our sample. The dependent variable in all regressions is an indicator of whether the loan has become delinquent after the origination. Our main explanatory variable is the interest rate on the loan at origination as well as its interaction with the fintech indicator. We also control for loan size and loan maturity, borrowers' credit score, age of credit history, number of accounts, and borrower age as well as zip code by origination year, month fixed effects. Standard errors are double clustered

Table 10
Repeat borrowers

Dep. var.	$I(Fintech_t) \times 100$					
	(1)	(2)	(3)	(4)	(5)	(6)
Fintech indicator for last loan	0.623*** (0.003)	0.621*** (0.003)	0.615*** (0.003)	0.556*** (0.003)	0.614*** (0.003)	0.609*** (0.003)
× Low credit				0.089*** (0.004)		
× Thin credit					0.094*** (0.011)	
× Main lender						0.272*** (0.007)
Male indicator		0.002*** (0.001)	0.003*** (0.001)	0.003*** (0.001)	0.003*** (0.001)	0.003*** (0.001)
Married indicator		0.001** (0.001)	-0.002*** (0.001)	-0.002*** (0.001)	-0.002*** (0.001)	-0.002*** (0.001)
College indicator		0.012*** (0.001)	0.010*** (0.001)	0.010*** (0.001)	0.010*** (0.001)	0.010*** (0.001)
High income indicator		0.019*** (0.001)	0.015*** (0.001)	0.016*** (0.001)	0.016*** (0.001)	0.015*** (0.001)
Young indicator		0.004*** (0.001)	0.004*** (0.001)	0.004*** (0.001)	0.004*** (0.001)	0.004*** (0.001)
Professional indicator		0.010*** (0.001)	0.008*** (0.001)	0.009*** (0.001)	0.008*** (0.001)	0.008*** (0.001)
Controls	No	No	Yes	Yes	Yes	Yes
Zip × Time FE	Yes	Yes	Yes	Yes	Yes	Yes
N	1,695,511	1,695,510	1,650,172	1,650,172	1,650,172	1,650,172
R ²	0.689	0.689	0.689	0.69	0.689	0.69

The table reports the regression results of likelihood of fintech borrowers taking another fintech loan in the future

(a) Table 9: Pricing and ex post performance. The table tests whether loan note rates predict later delinquency and whether fintech rates are more informative about default risk.

(b) Table 10: Repeat borrowers. The table estimates whether previous fintech borrowing predicts future fintech borrowing, controlling for borrower and local-market characteristics.

Table 11
Job loss and fintech loans

Dep. var.	I(Fintech _{i,t}) × 100		I(Non-fintech _{i,t}) × 100		I(Any Loan _{i,t}) × 100	
Sample	Fintech borrowers		Non-fintech borrowers		All	
	(1)	(2)	(3)	(4)	(5)	(6)
Post job loss _t	0.404** (0.104)	0.260*** (0.109)	0.138** (0.061)	−0.154*** (0.065)	0.097 (0.061)	−0.216*** (0.065)
× Fintech _i					0.397*** (0.121)	0.656*** (0.127)
Fintech _i × Unemployed _i					−0.329*** (0.089)	
Unemployed _i	0.020** (0.075)		0.210*** (0.049)		0.247*** (0.049)	
County time FE	Yes	No	Yes	No	Yes	No
Borrower FE	No	Yes	No	Yes	No	Yes
Calendar time FE	Yes	Yes	Yes	Yes	Yes	Yes
<i>N</i>	6,935,265	6,935,265	23,622,976	23,622,976	30,558,241	30,558,241
<i>R</i> ²	0.031	0.053	0.023	0.058	0.021	0.057

The table reports the regression results examining the relationship between job loss and personal loan origination. The regression is based on our loan-month panel data matched with individual job loss data. We restrict the individuals with job loss to three months before and three months after their job loss date. Those with no job loss are the same as our loan-month panel. The dependent variable is an indicator for whether the loan (fintech, non-fintech or both) is originated in a given month. *Post job loss* is an indicator that takes 1 for individuals with job loss after their job loss date. Unemployment is an indicator for the individuals with job loss during our sample period. Regressions in columns (1) and (2) are based on fintech borrowers only. Regressions in columns (3) and

Table 11: Job loss and fintech loans. The table relates employment shocks to subsequent fintech and non-fintech borrowing, showing that fintech credit is especially relevant when borrowers face liquidity needs after job loss.

7 Short summary

- The paper studies whether fintech lenders improve consumer credit allocation or simply take risky borrowers.
- It uses credit-bureau data that identify lender names and track borrower credit outcomes monthly.
- Fintech lenders grow rapidly from 2012 to 2017.
- Fintech lenders initially serve riskier borrowers and later move toward safer borrowers.
- Fintech borrowers typically have mid-range credit scores, many accounts, and high utilization.
- Fintech pricing relies heavily on hard credit-report information.
- Fintech loans are more likely to default than comparable non-fintech loans.
- This result survives zip-time fixed effects, rich controls, matching, entropy balancing, and borrower fixed effects.
- Borrowers initially consolidate debt but later increase total and revolving debt.
- Fintech rates predict future default strongly, suggesting that lenders price risk on the intensive margin.
- Repeat borrowing is common, which may raise customer lifetime value for fintech lenders.
- The paper’s main interpretation is that fintech lending expands access and convenience but may miss soft-information risks and encourage additional borrowing.

8 Conclusion

8.1 Core conclusion

Di Maggio and Yao show that fintech lenders are neither pure cream-skimmers nor simply benevolent providers of credit to excluded borrowers. They expand rapidly by serving borrowers who often already have access to credit, initially with riskier borrower profiles and later with safer borrowers. Their borrowers default more than similar traditional-loan borrowers, but the lenders’ pricing models appear to recognize some of that risk.

8.2 Incremental conclusion for fintech lending

The paper’s incremental contribution is to connect the fintech lending debate to ex post borrower balance-sheet dynamics. It shows that the key issue is not only whether fintech lenders approve borrowers faster or cheaper, but what borrowers do after receiving the loan. The initial debt-consolidation benefit can be short-lived if the borrower rebuilds revolving debt and finances new consumption.

8.3 Economic mechanism

The evidence supports a mechanism with three linked parts:

1. fintech lenders use hard credit data to price and approve loans quickly;
2. some borrowers attracted to fintech credit carry soft-information or behavioral risks not fully captured by standard credit variables;
3. easier access to liquidity can increase consumption and subsequent indebtedness, raising delinquency risk.

8.4 Policy implications

The results imply that fintech entry can increase competition and expand credit availability, but it can also raise borrower fragility. Regulators and researchers should evaluate fintech lending not only by approval rates and interest rates but also by post-origination borrower outcomes, including total debt, credit score dynamics, and delinquency across all accounts.

8.5 Limitations and open questions

The paper cannot fully separate adverse selection from moral hazard because borrowers choose lender types and may face time-varying unobserved shocks. The data do not observe all fees, non-credit-bureau soft information, or the full lender profit function. Future research could study lender-level funding, platform algorithms, borrower welfare, and regulatory changes that affect fintech underwriting.

8.6 Takeaway

The paper reframes fintech credit as a tradeoff. Fintech technology makes credit faster and more scalable, and it may price hard-information risk well. But speed and standardization do not necessarily improve borrower welfare when borrower risk is partly soft, behavioral, or time-varying. The central lesson is that financial innovation must be evaluated using both lender pricing and borrower outcomes.